Q1

The rms speed of oxygen at room temperature is about $500\ \text{ms}^{-1}$. The rms speed of hydrogen at the same temperature is about

(1) $125\ \text{m s}^{-1}$

(2) $2000\ \text{m s}^{-1}$

(3) $8000\ \text{m s}^{-1}$

(4) $500\ \text{m s}^{-1}$

Q2

One drop of soap bubble of diameter D breaks into 27 drops having surface tension T. The change in surface energy is

(1) $2\pi TD^2$

(2) $4\pi TD^2$

(3) πTD^2

(4) $8\pi TD^2$

Q3

Consider a monochromatic linear source of light which is placed at a distance of r from a very small metal plate. Due to photoelectric effect electrons are ejected at rate N per second and it is found that maximum kinetic energy of photoelectron is K. If source is brought closer to a distance $\frac{r}{3}$, the rate of emission and the maximum kinetic energy of photoelectron become nearly

(1) $\frac{N}{3}, 3K$

(2) $3N, 3K$

(3) $3N, K$

(4) $3N, \frac{K}{3}$

Q4

The respective number of significant figures for the numbers 46.035, 0.0007 and 4.1×10^{-3} are

(1) $5, 1, 2$

(2) $5, 1, 5$

(3) $5, 5, 2$

(4) $4, 4, 2$

Q5

A conducting hollow spherical shell having an inner radius a and outer radius b carries a net charge Q. If a point charge q is placed at the centre of this sphere, the surface charge density of inner and outer surface will be -

(1) $\frac{-q}{4\pi a^2}, \frac{-q+Q}{4\pi b^2}$

(2) $\frac{q}{4\pi a^2}, \frac{q+Q}{4\pi b^2}$

(3) $\frac{-q}{4\pi a^2}, \frac{q+Q}{4\pi b^2}$

(4) 0,0

Q6

Assertion: In series combination of electric bulbs, bulb of lower power emits more light than higher power bulb.

Reason: The lower power bulb in series gets more current than the higher power bulb.

(1) Both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

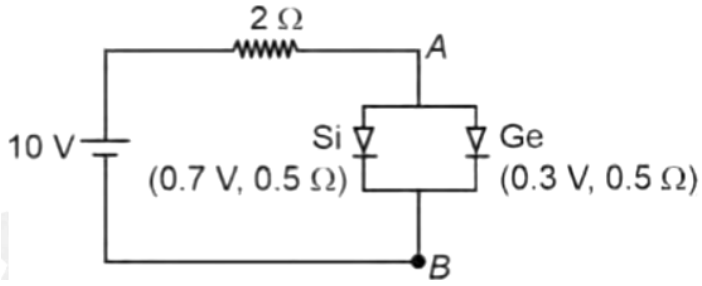
(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but the Reason is false.

(4) Assertion is false but the Reason is true.

Q7

The current through the battery in the given circuit will be



(1) $\frac{38}{9}$ A

(2) $\frac{37}{7}$ A

(3) 4 A

(4) 5 A

Q8

An ammeter is obtained by shunting a 30Ω galvanometer with a 30Ω resistance. The additional shunt required to double the range is

(1) 15Ω

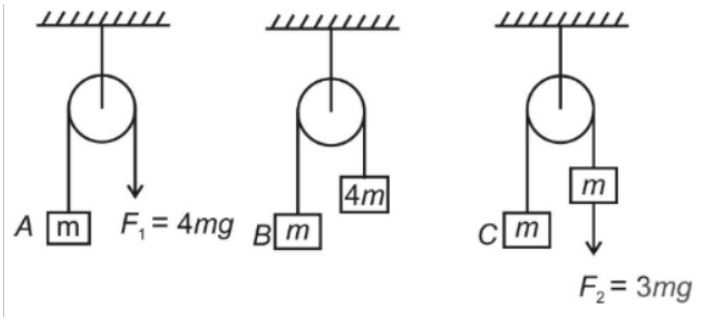
(2) 10Ω

(3) 5Ω

(4) 30Ω

Q9

In the figure, three blocks A, B and C of mass m each, have acceleration a_1, a_2 and a_3 respectively. F_1 and F_2 are external forces of magnitude $4mg$ and $3mg$ respectively then



(1) $a_1 > a_2 > a_3$

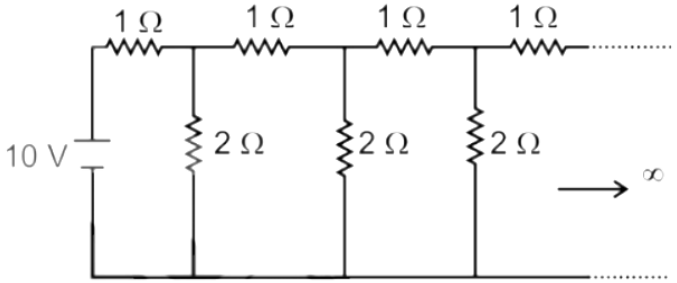
(2) $a_1 = a_2 = a_3$

(3) $a_1 > a_3 > a_2$

(4) $a_1 < a_3 < a_2$

Q10

The current flowing through the battery connected with infinite network as shown in the figure is



(1) 5 A

(2) 10 A

(3) 2 A

(4) 4 A

Q11

The rms value of electric field at a distance r from a source of electromagnetic wave of power P is (ε : Permittivity of free space)

(1) $\sqrt{\frac{P}{2\pi r^2 c\varepsilon}}$

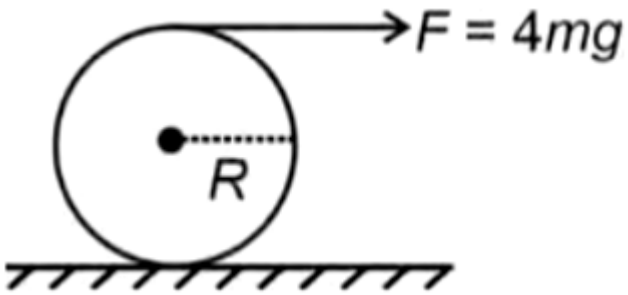
(2) $\sqrt{\frac{P}{4\pi r^2 c\varepsilon}}$

(3) $\sqrt{\frac{P}{8\pi r^2 c\varepsilon}}$

(4) $\sqrt{\frac{2P}{\pi r^2 c\varepsilon}}$

Q12

A solid sphere of mass m and radius R is placed on a rough horizontal surface. It is pulled by a horizontal force F acting at the top of the sphere as shown in figure. If sphere rolls purely over the surface then acceleration of centre of mass of the sphere is



(1) $\frac{20g}{7}$

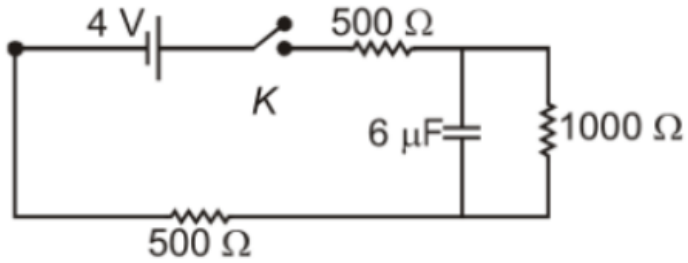
(2) $\frac{30g}{7}$

(3) $\frac{40g}{7}$

(4) $\frac{50g}{7}$

Q13

In the circuit shown below, if the tap key K is pressed at time $t = 0$, then which of the following statements is false?



(1) At time $t = 0$; current delivered by battery $= 4\ \text{mA}$

(2) At time $t = \infty$; current delivered by battery $= 2\ \text{mA}$

(3) At time $t = \infty$; charge on capacitor $= 6\mu\text{C}$

(4) At time $t = \infty$; charge on capacitor $= 12\mu\text{C}$

Q14

A block of mass 2 kg is initially at rest on a horizontal frictionless surface. A horizontal force $\overrightarrow{\mathbf{F}} = \left(9 - x^2\right) \hat{\mathbf{i}}$ newtons acts on it, when the block is at $x = 0$. The maximum kinetic energy of the block between $x = 0$ and $x = 3$ m in joule is

(1) 24

(2) 20

(3) 18

(4) 15

Q15

A transformer is used on a 2000 V input line to deliver 20 A at 120 V at the secondary coil. If the power efficiency of the transformer is 80% then the current drawn from the input line is

(1) 3 A

(2) 2 A

(3) 1.5 A

(4) 1.25 A

Q16

Which of the following statements are true regarding Bohr's model of hydrogen atom?

(I) Orbiting speed of electron decreases as it shifts to discrete orbits away from the nucleus.

(II) Radii of allowed orbits of electron are proportional to the principal quantum number.

(III) Frequency with which electrons orbit around the nucleus in discrete orbits is inversely proportional to the cube of principal quantum number.

(IV) Binding force with which the electron is bound to the nucleus increases as it shifts to outer orbits.

Select the correct answer using the codes given below

(1) I and III

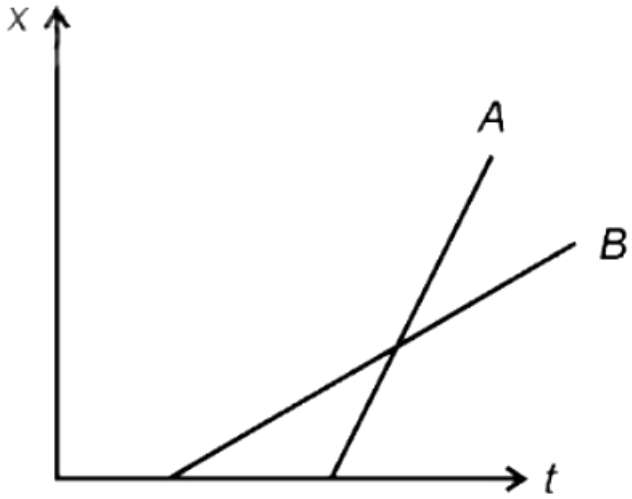
(2) II and IV

(3) I, II and III

(4) II, III and IV

Q17

Figure shows position (x)-time (t) graph for two boys going from their homes to school. Which of the following statement is true about their relative velocity, when both are in motion?



(1) First increase then decrease

(2) First decrease then increase

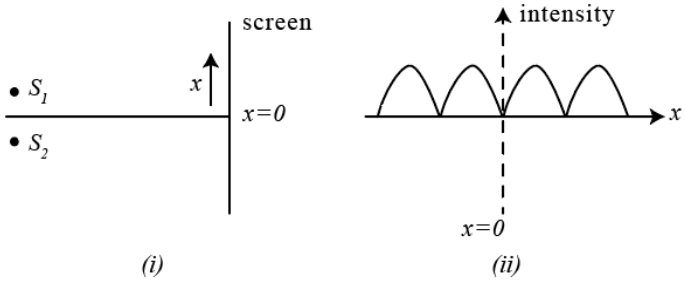
(3) Is zero

(4) Is non zero but constant

Q18

Two light sources S_1 and S_2 of intensity I_1 and I_2 respectively are placed in front of a screen as shown in figure (i). The pattern of intensity distribution observed in central portion is shown in figure (ii). If the intensity

distribution is found to be constant with time, then choose the incorrect statement.



(1) S_1 and S_2 have the same intensities

(2) S_1 and S_2 have constant initial phase difference

(3) S_1 and S_2 have same frequency

(4) Phase difference between the two waves meeting at $x = 0$ is zero.

Q19

A monoatomic gas is compressed from a volume of $2\ \mathrm{m}^3$ to a volume of $1\ \mathrm{m}^3$ at a constant pressure of $100\mathrm{Nm}^{-2}$. Then it is heated at constant volume by supplying 150 J of energy. As a result, the internal energy of the gas

(1) increase by 250 J

(2) decrease by 250 J

(3) decrease by 50 J

(4) remains same

Q20

Imagine a light planet revolving around a very massive star in a circular orbit of radius R with time period of revolution T. If the gravitational force of attraction between the planet and the star is proportional to $R^{-\frac{3}{2}}$, then

(1) T is proportional to $R^{\frac{5}{4}}$

(2) T is proportional to $R^{\frac{7}{4}}$

(3) T is proportional to $R^{\frac{3}{2}}$

(4) T is proportional to R

Q21

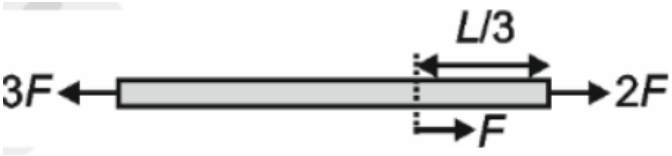
A bar magnet has a magnetic moment of $200\mathrm{Am}^2$. The magnet is suspended in a magnetic field of $0.30\mathrm{NA}^{-1}\mathrm{m}^{-1}$. Torque required when it is at angle 30 degree from its equilibrium will be (in Nm)

Q22

If the rate of change in velocity with respect to time is constant and its position after 6th second will be same as that after 11th second then the particle returns to the starting point at time t equals to

Q23

A uniform cylindrical rod of length L, cross sectional area A and Young's modulus Y is acted upon by the forces shown in the figure. The elongation of the rod is $\frac{nFL}{3AY}$, find n.

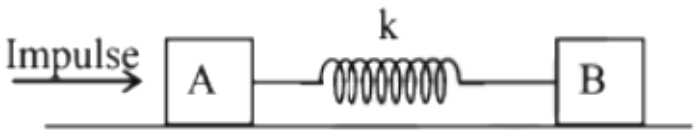


Q24

In YDSE intensities at two points P_1 and P_2 on the screen are I_1 and I_2 respectively. If P_1 is located at the centre of a bright fringe and P_2 is located at a distance equal to a $1/4th$ of fringe width from P_1, then $\frac{\mathrm{I}_1}{\mathrm{I}_2}$ is

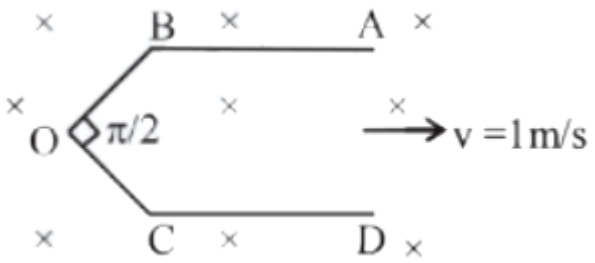
Q25

Two blocks A and B of masses 8 kg and 4 kg placed on a smooth horizontal surfaces, are connected with a spring. An impulse of 48 N-s provides to A. Find the velocity of centre of mass in (m/sec) of System just after applying impulse $(\mathrm{k} = 100\ \mathrm{N/m})$



Q26

A conductor ABOCD moves along its bisector with a velocity 1 m/s through a perpendicular magnetic field of 1 T, as shown in figure. If all the four sides are 1 m length each, then the induced emf between A and D is a $\sqrt{2}$ volt, then "a" is -

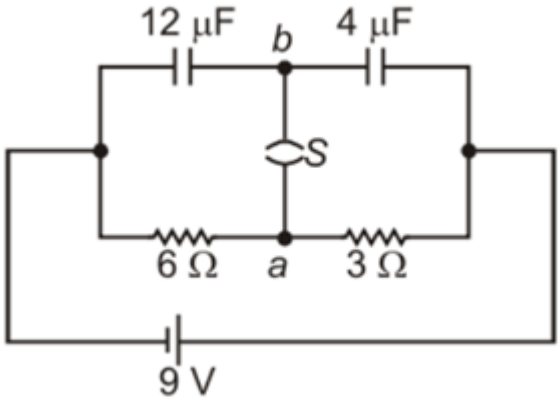


Q27

A hydrogen like atom has one electron revolving round a stationary nucleus. If the energy required to excite the electron from the 2 nd orbit to 3 rd orbit is 47.2eV, find the atomic number of the given atom. (Mark answer to nearest integer)

Q28

For the circuit shown below, when the switch S is closed, then the total amount of charge that flows from point a to point b through the switch, is (mark answer in micro Coulomb)

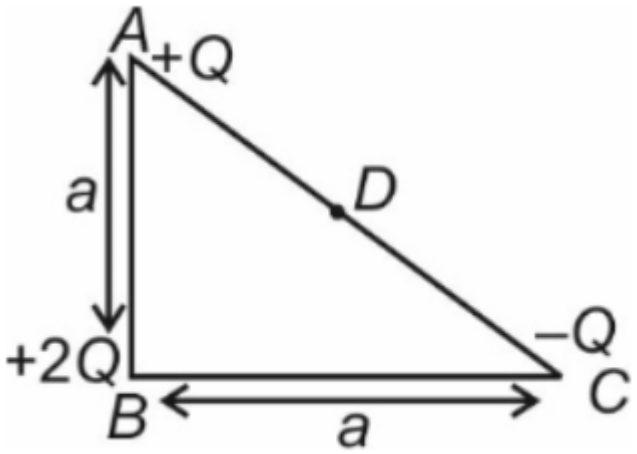


Q29

A particle is executing simple harmonic motion with an amplitude A and time period T. The displacement of the particles after $2T$ period from its initial position is _____ A.

Q30

Three charges are placed (fixed) at the vertices of an isosceles right angle triangle ABC with point charge $+Q, +2Q \& -Q$ are placed at points $A, B \& C$ respectively. The magnitude of electric field strength at point D (mid point of AC) is $\sqrt{\eta}\,\frac{KQ}{a^2}$. Find η.



Q31

Column I (Species)	Column II (Colour in aq - solution)
a) Co^{3+}, Co^{2+}	p) Yellow
b) Fe^{2+}	q) Blue pink
c) Cu^{2+}	r) Green
d) Mn^{3+}	s) Violet
e) Fe^{3+}	t) Blue

(1) a-q, b-r, c-t, d-s, e-p

(2) a-t, b-r, c-q, d-s, e-p

(3) a-p, b-r, c-q, d-s, e-t

(4) a-q, b-s, c-p, d-r, e-t

Q32

Select the correct order of metallic character :

(1) $P < Si < Be < Mg < Na$

(2) $Be < P < Si < Mg < Na$

(3) $P < Be < Si < Na < Mg$

(4) $Si < P < Be < Mg < Na$

Q33

Consider the following statements

Statement 1: $\left[Cr(NH_3)_6\right]^{3+}$ is an inner orbital complex with crystal field stabilization energy equal to $-1.2\Delta_0$

Statement 2: The octahedral complex formed by joining the CN^-ligands to Fe^{3+} ion has theoretical value of spin only magnetic moment equal to $1.73BM$

Statement 3: $Na_2\ S + Na_2\left[Fe(CN)_5NO\right] \rightarrow Na_4\left[Fe(CN)_5NOS\right]$, In reactant and product the oxidation states of iron are same.

Arrange in the order of true / false

(1) FTF

(2) TTF

(3) TTT

(4) FFF

Q34

Which of the following statements is not correct?

(1) The first ionization energies (in $KJmol^{-1}$) of carbon, silicon, germanium, tin, and lead are 1086, 786, 761, 708 and 715 respectively

(2) Down the group, ionization energy decreases regular from B to Tl in boron family

(3) Among oxides of the elements of carbon family, CO is neutral, GeO is acidic and SnO is amphoteric

(4) The 4f - and 5f-inner transition elements are placed separately at the bottom of the periodic table

Q35

Assertion : Water is a good solvent for ionic compounds but poor one for covalent compounds.

Reason : Hydration energy of ions releases sufficient energy to overcome lattice energy and break hydrogen

bonds in water, while covalent bonded compounds interact so weakly that even Vander Wall's forces between molecules of covalent compounds cannot be broken.

(1) both assertion and reason are true and the reason is the correct explanation of the assertion.

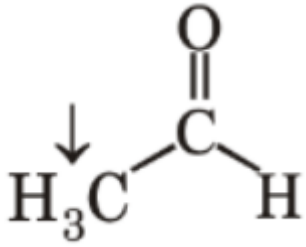
(2) both assertion and reason are true but reason is not the correct explanation of the assertion.

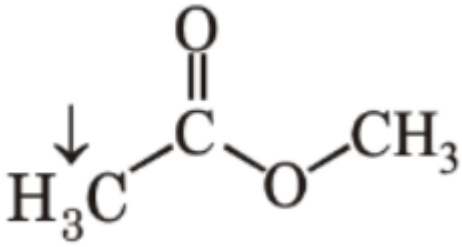
(3) assertion is true but reason is false.

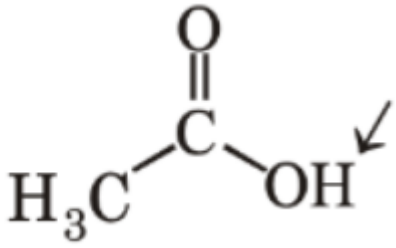
(4) the assertion and reason both are false.

Q36

The relative acidity of the indicated H in each of the following is

(i) 

(ii) 

(iii) 

(1) i > ii > iii

(2) ii > iii > i

(3) i > iii > ii

(4) iii > i > ii

Q37

In a hydrogen like species, an electron jumps from an orbital having two radial and two angular nodes to the orbital having same sign of wave function in all direction, at any distance. If the energy of emitted photon is 326.4eV, the atom or ion is

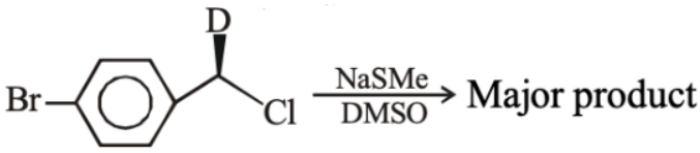
(1) H

(2) Li^{2+}

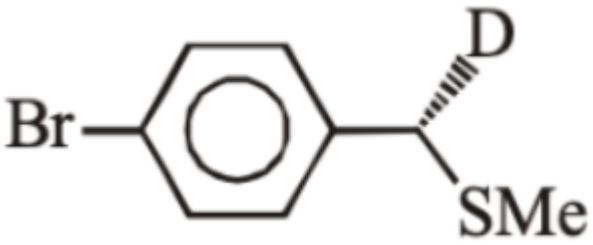
(3) He^{+}

(4) B^{4+}

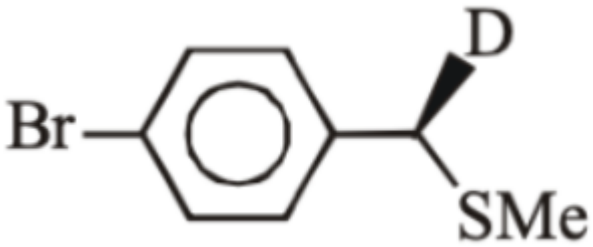
Q38



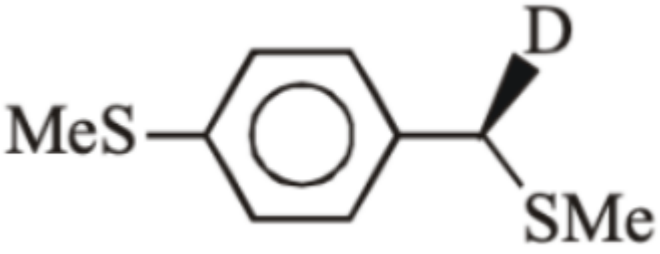
(1)

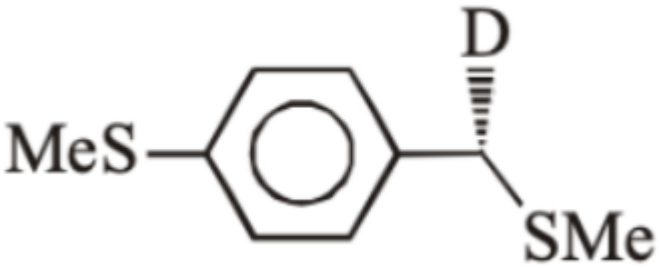


(2)



(3)





(4)

Q39

For a complex $\left[\mathrm{Co(NH_3)_3Cl_3}\right]$ pick up true statements?

(1) The co-ordination number and oxidation number of cobalt is 6

(2) The complex can show fac and mer isomerism

(3) The complex can show optical isomerism

(4) This complex can form white precipitate of AgCl with $\mathrm{AgNO_3}$

Q40

Select correct order :

(1) $s-s > s-p > p-p$ (order of extent of overlapping)

(2) $\mathrm{CO_3^{-2}} > \mathrm{CO_2} > \mathrm{CO}$ (order of $\mathrm{C-O}$ bond length)

(3) $\mathrm{PF_3} < \mathrm{BF_3} < \mathrm{ClF_3}$ (order of bond angle)

(4) $\mathrm{HF} < \mathrm{NH_3} < \mathrm{H_2\,S}$ (order of boiling point)

Q41

4 mol of a mixture of Mohr's salt and $\mathrm{Fe_2(SO_4)_3}$ requires $500\mathrm{ml}$ of $1\mathrm{M}, \mathrm{K_2Cr_2O_7}$ for complete oxidation in acidic medium. The mole $\%$ of the Mohr's salt in the mixture is -

(1) 25

(2) 50

(3) 60

(4) 75

Q42

Which of the options contain correct set of parameters which are both amount and path independent parameters.

(I) Internal energy

(II) Specific enthalpy

(III) Heat exchange at constant pressure condition

(IV) Molar heat exchange at constant volume condition

(1) Only II, IV

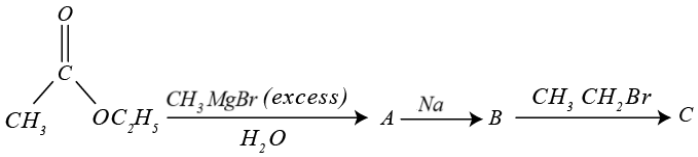
(2) All option

(3) All options except III

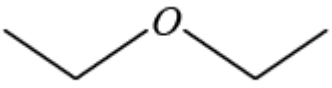
(4) Only II

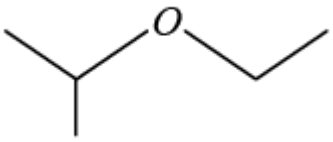
Q43

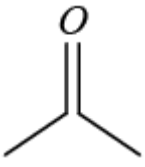
Consider the following sequence of reactions.

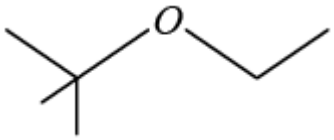


The final product (C) is -

(1) 

(2) 

(3) 



(4)

Q44

The correct order of ionic radii of $\mathrm{Yb}^{3+}, \mathrm{La}^{3+}, \mathrm{Eu}^{3+}$ and Lu^{3+} is

(Atomic No. $Yb = 70, La = 57, Eu = 63, Lu = 71$)

(1) $\mathrm{La}^{3+} < \mathrm{Eu}^{3+} < \mathrm{Lu}^{3+} < \mathrm{Yb}^{3+}$

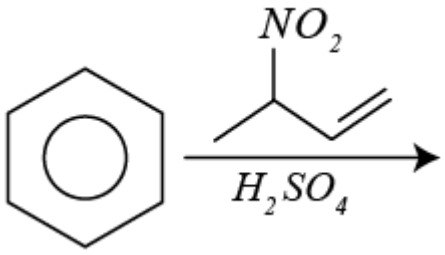
(2) $\mathrm{Yb}^{3+} < \mathrm{La}^{3+} < \mathrm{Eu}^{3+} < \mathrm{Lu}^{3+}$

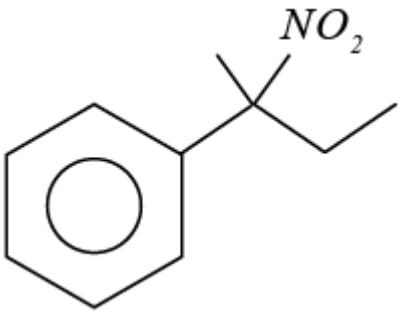
(3) $\mathrm{Lu}^{3+} < \mathrm{Yb}^{3+} < \mathrm{Eu}^{3+} < \mathrm{La}^{3+}$

(4) $\mathrm{Lu}^{3+} < \mathrm{Eu}^{3+} < \mathrm{La}^{3+} < \mathrm{Yb}^{3+}$

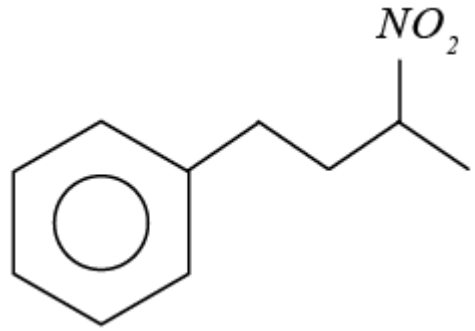
Q45

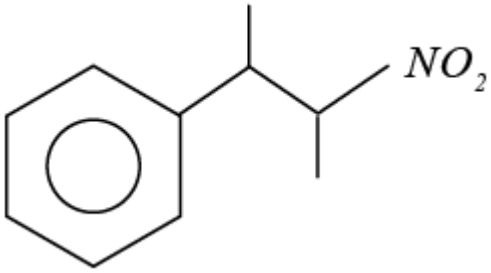
The major product of the following reaction is:

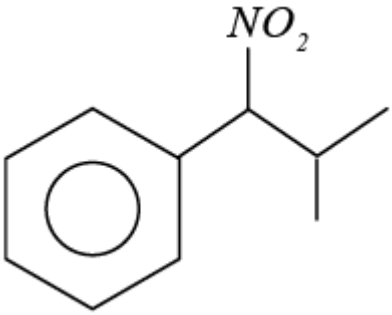




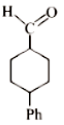
(1)

(2) 

(3) 

(4) 

Q46

List-I (Compound)	List-II (Statements)
(P) 	(1) It can show tautomerism, optical isomerism but not geometrical isomerism
(Q) 	(2) It can show tautomerism, geometrical isomerism but not optical isomerism
(R) 	(3) It can show geometrical isomerism, optical isomerism but not tautomerism
(S) 	(4) It can show tautomerism, geometrical isomerism & also optical isomerism

(1)

P	Q	R	S
2	1	3	4

(2)

P	Q	R	S
2	3	4	1

(3)

P	Q	R	S
3	2	4	1

(4)

P	Q	R	S
2	3	1	4

Q47

Statement 1: Oxygen is more electronegative than sulphur, yet H_2S is more acidic than H_2O.

Statement 2: $H-S$ bond is weaker than $O-H$ bond.

(1) Statement 1 is True, Statement 2 is true; Statement 2 is a correct explanation for Statement 1

(2) Statement 1 is True, Statement 2 is true; Statement 2 is NOT a correct explanation for Statement 1

(3) Statement I is True, Statement 2 is False

(4) Statement 1 is False, Statement is True

Q48

Let solubility of a sparingly soluble salt of strong base and weak acid in pure water is equal to s_1, in a buffer solution with pH < 7 is equal to s_2 and in another buffer solution with pH > 7 is equal to s_3. If hydrolysis is considered in all the solutions then the correct relation between s_1, s_2 and s_3 is

(1) $s_2 = s_2 = s_3$

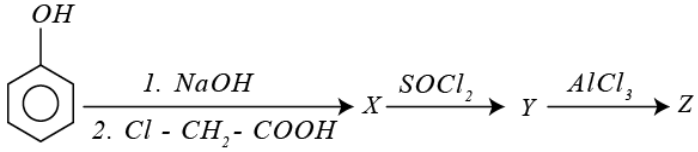
(2) $s_1 > s_2 > s_3$

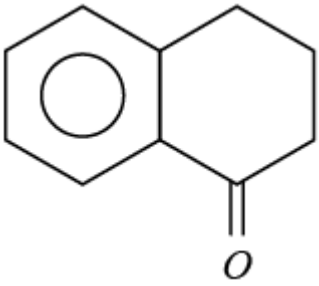
(3) $s_1 < s_2 < s_3$

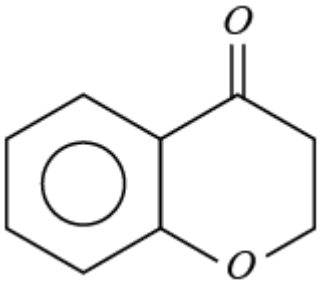
(4) $s_2 > s_1 > s_3$

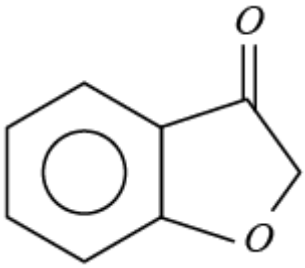
Q49

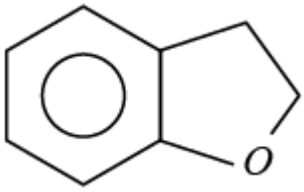
Identify ' Z ' in the given sequence of reaction.



(1) 

(2) 

(3) 

(4) 

Q50

Correct order of bond angle is -

(1) $PH_4^+ > OF_2 > SF_2 > SbH_3$

(2) $OF_2 > SF_2 > PH_4^+ > SbH_3$

(3) $PH_4^+ > SF_2 > OF_2 > SbH_3$

(4) $SF_2 > OF_2 > PH_4^+ > SbH_3$

Q51

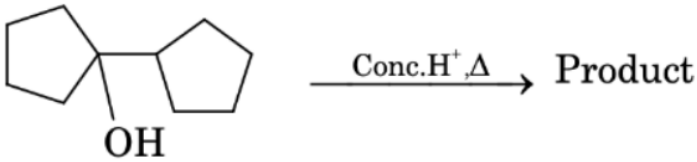
For the estimation of nitrogen, 1.4 g of an organic compound was digested by Kjeldahl method and the evolved ammonia was absorbed in 60 mL of $\frac{\mathrm{M}}{10}$ sulphuric acid. The unreacted acid required 20 mL of $\frac{\mathrm{M}}{10}$ sodium hydroxide for complete neutralizaton. The percentage of nitrogen in the compound is :

Q52

At the given condition of $CH_4 - O_2$ fuel cell the cell emf is 0.8 V and the enthalpy of combustion of $CH_4(g)$ is $-772\ \mathrm{kJ/mol}$. The maximum efficiency of the given fuel cell in the given condition is _____ %

Q53

How many transition states are formed during the formation of product (major)



Q54

For the reaction : $xA \longrightarrow yB$,

$\log_{10}\left(-\frac{\mathrm{d[A]}}{\mathrm{dt}}\right) = \log_{10}\left(+\frac{\mathrm{d[B]}}{\mathrm{dt}}\right) + 0.3$

If the value of $\log_{10} 2 = 0.3$, the value of $x : y$ is

Q55

All the energy released from the reaction $\mathrm{X} \rightarrow \mathrm{Y}$, $\Delta_r G^0 = -193\ \mathrm{kJ\ mol^{-1}}$ is used for oxidizing $\mathrm{M^+}$as $\mathrm{M^+ \rightarrow M^{3+} + 2}e^-$, $E^0 = -0.25\ \mathrm{V}$.

Under standard conditions, the number of moles of $\mathrm{M^+}$oxidized when one mole of X is converted to Y is _____. $\left[F = 96500\mathrm{C mol^{-1}}\right]$

Q56

How many of the following reactions are disproportionation reactions

(a) $\mathrm{Cu}^{+} \xrightarrow[\text{medium}]{\text{aqueous}} \mathrm{Cu}^{2+}(\mathrm{aq}) + \mathrm{Cu}$

(b) $3\mathrm{MnO_4}^{2-} + 4\mathrm{H}^{+} \longrightarrow 2\mathrm{MnO_4}^{-} + \mathrm{MnO_2} + 2\mathrm{H_2O}$

(c) $2\mathrm{KMnO_4} \xrightarrow{\Delta} \mathrm{K_2MnO_4} + \mathrm{MnO_2} + \mathrm{O_2}$

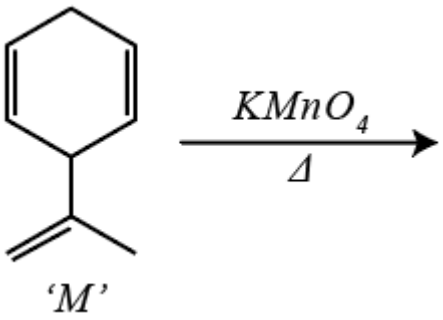
(d) $2\mathrm{MnO_4^{-}} + 3\mathrm{Mn}^{2+} + 2\mathrm{H_2O} \longrightarrow 5\mathrm{MnO_2} + 4\mathrm{H}^{+}$

Q57

If 1gm of solute dimerise upto 75% in 100gm of $\mathrm{H_2O}$ & show depression in freezing point $0.093^{\circ}\mathrm{C}$. Find average molecular weight of solute in the solution $\left(\mathrm{K_{f(H_2O)}} = 1.86\right)$.

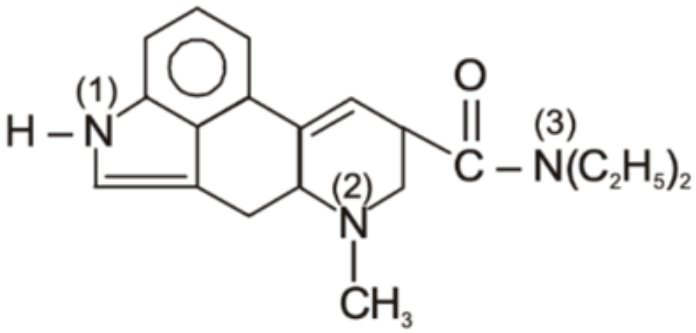
Q58

Total number of moles of CO_2 evolved in following reaction per mole of compound ' $\mathbf{M}$ ' is

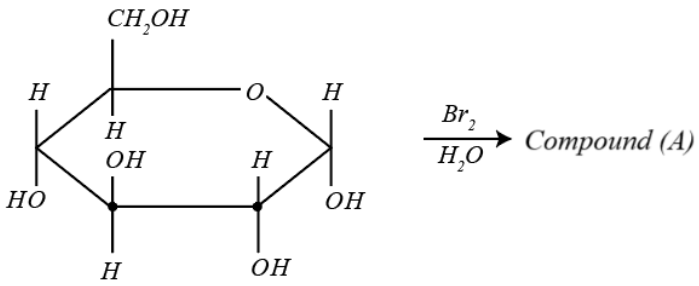


Q59

Which nitrogen in (Lysergic acid diethylamide) is most basic ?



Q60



Number of chiral centre in compound A is

Q61

Set P consists of m consecutive integers whose sum is 2 m. Set Q consists of 2 m consecutive integers whose sum is m. The difference between greatest element of set P & set Q is 1009 then m is

(1) 2017

(2) 2019

(3) 2021

(4) 2023

Q62

If $\frac{\sin x}{\sin y}=\frac{1}{2}, \frac{\cos x}{\cos y}=\frac{3}{2}$, where $x, y, \in\left(0, \frac{\pi}{2}\right)$, then the value of $\tan(x+y)$ is equal to

(1) $\sqrt{13}$

(2) $\sqrt{14}$

(3) $\sqrt{17}$

(4) $\sqrt{15}$

Q63

Let $a_1, a_2, \ldots \ldots a_{101}$ is a group of real numbers such that $a_1 > a_{i+1}$ for all values of i and mean square deviation of the group is minimum about the number ' a_{51} ', then mode of the group will be -

(1) $2a_{51}$

(2) a_{51}

(3) a_{50}

(4) a_{52}

Q64

If $f : \mathrm{R} \to \mathrm{R}$ is continuous and differentiable function such that

$\int_{-1}^{\mathrm{x}} f(\mathrm{t})\mathrm{dt} + f'''(3)\int_{\mathrm{x}}^{0} \mathrm{dt} = \int_{1}^{\mathrm{x}} \mathrm{t}^3\mathrm{dt} - f'(1)\int_{0}^{\mathrm{x}} \mathrm{t}^2\mathrm{dt} + f''(2)\int_{\mathrm{x}}^{3} \mathrm{t}dt$ then the value of $f'(4)$ is

(1) $48 - 8f'(1) + f''(2)$

(2) $48 - 8f'(1) - f''(2)$

(3) $48 + 8f'(1) + f''(2)$

(4) 0

Q65

Let r be a relation from R (Set of real number) to R defined by $r = \{(x, y) \mid x, y \in R$ and xy is an irrational number $\}$, then relation r is -

(1) reflexive and symmetric only

(2) symmetric only

(3) symmetric and transitive only

(4) equivalence relation

Q66

A man alternatively tosses a coin and throws a die, beginning with tossing the coin. The probability that he gets a tail on the coin before he gets a 3 or 4 on the die is

(1) $\frac{1}{3}$

(2) $\frac{1}{2}$

(3) $\frac{2}{3}$

(4) $\frac{3}{4}$

Q67

If $\int \frac{\sec x(2+\sec x)}{(1+2\sec x)^2}dx = f(x) + C$, where C is constant of integration and $f(0) = 0$, then $\int_0^{\pi} f(x)dx$ is equal to

(1) 0

(2) $\ln 2$

(3) $\ln 3$

(4) $\ln 4$

Q68

Let $g(x) = 6x^2 - 18x + 8$;

$f_1(\mathrm{x}) = |\mathrm{g(x)}|, f_2(\mathrm{x}) = |f_1(\mathrm{x}) - \mathrm{P}_1|, f_3(\mathrm{x}) = |f_2(\mathrm{x}) - \mathrm{P}_2|$.

If $\mathrm{P}_1 = 7$, then the largest integral value of P_2 such that $f_3(\mathrm{x})$ has exactly 10 points of non-differentiability, is

(1) 6

(2) 4

(3) 5

(4) 7

Q69

The solution of the equation $(y\ln x - 1)ydx - xdy = 0$ is _____.

(1) $y(1 - \ln x + cx) = 1$

(2) $x(1 + \ln x + cy) = 1$

(3) $y(1 + \ln x + cx) = 1$

(4) none of these

Q70

The two lines whose direction cosines are connected by the relations $\mathrm{a}\ell + \mathrm{bm} + \mathrm{cn} = 0$ and

$\mathrm{u}\ell^2 + \mathrm{vm}^2 + \mathrm{wn}^2 = 0$ are parallel then

(1) $a^2(v - w) + b^2(w - u) + c^2(u - v) = 0$

(2) $\frac{a^2}{u}+\frac{b^2}{v}+\frac{c^2}{w}=0$

(3) $a\left(v^2-w^2\right)+b^2\left(w^2-u^2\right)+c\left(u^2-v^2\right)=0$

(4) $a^2(v+w)+b^2(w+u)+c^2(u+v)=0$

Q71

The number of 3×3 skew symmetric matrices A whose entries are chosen from $\{-1,0,1\}$ and for which the system $A\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$ has infinite solution is

(1) 27

(2) 8

(3) 9

(4) 4

Q72

If complex number z satisfies $(z-\bar{z})^2=12|z|^2-4$ then find the maximum value of $3\sqrt{3}\,\mathrm{Re}(z)+8\,\mathrm{Im}(z)$.

(1) 4

(2) 2

(3) 5

(4) 7

Q73

A circle has the same centre as an ellipse and passes through the foci F_1 and F_2 of the ellipse, such that the two curves intersect in four points. Let P be any one of their points of intersection. If the major axis of the ellipse is 17 and the area of the triangle PF_1F_2 is 30 , then the distance between the foci is

(1) 11

(2) 12

(3) 13

(4) 15

Q74

The remainder when $(2024)^{2024}$ is divided by 49 is K. Find the sum of digits of K.

(1) 2

(2) 7

(3) 5

(4) 6

Q75

Let $f(x)$ and $g(x)$ are two functions defined from $\mathrm{R}^{+} \rightarrow \mathrm{R}$ such that

$f(x)=\begin{cases}1-\sqrt{x}, & x \text{ is rational} \\ x^2, & x \text{ is irrational}\end{cases}$

and $g(x)=\begin{cases}x, & x \text{ is rational} \\ 1-x, & x \text{ is irrational}\end{cases}$, then the composite function. $\mathrm{f(g(x))}$ is

(1) one - one onto

(2) many - one into

(3) one - one into

(4) many - one onto

Q76

The value of $\frac{[\lim_{n\to\infty} n\tan(4\pi e n!)]}{2}$ is ([.] means greatest integer function)'

(1) does not exist

(2) 3

(3) 5

(4) 6

Q77

Let $P(\alpha, \beta)$ be a variable point which moves in x-y plane such that $\frac{PA}{PB} = 2$, where $A(1, 0)$ and $\mathrm{B}(0, -1)$. If M and m denote respectively the maximum and minimum value of $\alpha + \beta$, then value of $\left[\frac{M}{m}\right]$ is(where [.] denotes the greatest integer function)

(1) -1

(2) -3

(3) 0

(4) 1

Q78

The shortest distance between the lines $\frac{x+1}{2} = \frac{y+1}{3} = \frac{z+1}{4}$ and $\frac{x+1}{3} = \frac{y}{4} = \frac{z}{5}$ is

(1) 1

(2) $\frac{1}{\sqrt{2}}$

(3) $\frac{1}{\sqrt{3}}$

(4) $\frac{1}{\sqrt{6}}$

Q79

The equation $(x + 2)(x + 3)(x + 8)(x + 12) = 4x^2$ has

(1) two roots rational and two roots irrational

(2) all roots positive and rational

(3) all roots negative and rational

(4) all root irrational

Q80

Let $\overrightarrow{\mathrm{OA}}$ and $\overrightarrow{\mathrm{OB}}$ be two sides of a triangle. The median $\overrightarrow{\mathrm{AM}}$ is perpendicular to angle bisector $\overrightarrow{\mathrm{OL}}$ and $|\overrightarrow{\mathrm{AM}}| : |\overrightarrow{\mathrm{OL}}| = 1 : 2$. The angle between $\overrightarrow{\mathrm{OA}}$ and $\overrightarrow{\mathrm{OB}}$ is

(1) $\cos^{-1}(4/5)$

(2) $\cos^{-1}(1/2)$

(3) $\cos^{-1}(3/5)$

(4) $\cos^{-1}(1/\sqrt{2})$

Q81

The minimum value of

$\left(\frac{3}{\mathrm{a}}+1\right)^2+\left(\frac{\mathrm{a}}{\mathrm{b}}+1\right)^2+\left(\frac{\mathrm{b}}{\mathrm{c}}+1\right)^2+(3\mathrm{c}+1)^2$

where $0<\mathrm{a},\mathrm{b},\mathrm{c}\leq 9$, is $\mathrm{p}+\mathrm{q}\sqrt{\mathrm{r}};\mathrm{p},\mathrm{q},\mathrm{r}\in\mathrm{I}$ and q,r are coprimes, then $(\mathrm{p}+\mathrm{q}+\mathrm{r})$ is equal to

Q82

Let $\vec{a}=-\hat{i}-\hat{k},\vec{b}=-\hat{i}+\hat{j}$ and $\vec{c}=\hat{i}+2\hat{j}+3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{r}\cdot\vec{a}=0$ then $\vec{r}\cdot\vec{b}=$ _____.

Q83

The area enclosed by $g(x),x=-3,x=5$ and x-axis where $g(x)$ is inverse of $f(x)=x^3+3x+1$ is K, find 4K.

Q84

Let the two lines $L_1\equiv\left(\int_0^{2t}\left(\frac{\sin x}{x}+1\right)dx\right)x+y=3t$ and $L_2:2tx+y=0$ intersect at a point A. If the x-coordinate of the point A as $\mathrm{t}\to 0$ is expressed as $\left(\frac{\mathrm{p}}{\mathrm{q}}\right)$ (where p and q are relatively prime), then $(p+q)$ is

Q85

If $\left(2xy-y^2-y\right)dx=\left(2xy+x-x^2\right)dy$ and $y(1)=1$, then the value of $12|y(-1)|$ is

Q86

Let A be $m\times m$ matrix with all elements equal to 1 such that $A^n=16^{17}A,m,n\in N$, find sum of possible values of n.

Q87

Given $f(x)=\begin{cases} x|x| & \text{for} \quad x \leq -1 \\ [x+1]+[1-x] & \text{for} \quad -1<x<1 \\ -x|x| & \text{for} \quad x \geq 1 \end{cases}$

where [.] denotes the greatest integer function. If $I=\int_{-2}^{2} f(x)dx$ then $|3I|=\ldots\ldots$

Q88

From point P on hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ in first quadrant, line parallel to x-axis & y-axis are drawn cutting hyperbola at Q & R respectively. Let A be the vertex of hyperbola in 1st quadrant. If Area $(\triangle \text{APQ})=2\text{Area}(\triangle \text{APR})$ then eccentric angle of P is $\frac{\text{a}\pi}{\text{b}}$ (where a & b are relatively prime integers). The value of $a+b$ is

Q89

If f and g are two real valued differentiable functions such that $f(x)=x^2+g'(0)x+g''(1)$ and $g(y)=y^2+f''(y)+f'(y)-2$ then

$f(3)+\frac{1}{g(2)}$ is equal to (Roundoff answer to nearest integer)

Q90

The number of the positive integer pairs (x,y) such that $\frac{1}{x}+\frac{1}{y}=\frac{1}{2007}$ where $x<y$ is

Answer Key

Q1 (2)	**Q2** (4)	**Q3** (3)	**Q4** (1)
Q5 (3)	**Q6** (3)	**Q7** (1)	**Q8** (1)
Q9 (3)	**Q10** (1)	**Q11** (2)	**Q12** (3)
Q13 (3)	**Q14** (3)	**Q15** (3)	**Q16** (1)
Q17 (4)	**Q18** (4)	**Q19** (4)	**Q20** (1)
Q21 (30)	**Q22** (17)	**Q23** (8)	**Q24** (2)
Q25 (4)	**Q26** (1)	**Q27** (5)	**Q28** (60)
Q29 (0)	**Q30** (32)	**Q31** (1)	**Q32** (1)
Q33 (3)	**Q34** (2)	**Q35** (1)	**Q36** (4)
Q37 (4)	**Q38** (1)	**Q39** (2)	**Q40** (2)
Q41 (4)	**Q42** (1)	**Q43** (4)	**Q44** (3)
Q45 (2)	**Q46** (2)	**Q47** (1)	**Q48** (4)
Q49 (3)	**Q50** (1)	**Q51** (10)	**Q52** (80)
Q53 (5)	**Q54** (2)	**Q55** (4)	**Q56** (2)
Q57 (125)	**Q58** (4)	**Q59** (2)	**Q60** (4)
Q61 (3)	**Q62** (4)	**Q63** (2)	**Q64** (2)
Q65 (2)	**Q66** (4)	**Q67** (3)	**Q68** (1)

Q69 (3)	**Q70** (2)	**Q71** (1)	**Q72** (3)
Q73 (3)	**Q74** (4)	**Q75** (2)	**Q76** (4)
Q77 (3)	**Q78** (4)	**Q79** (1)	**Q80** (1)
Q81 (27)	**Q82** (9)	**Q83** (18)	**Q84** (5)
Q85 (12)	**Q86** (132)	**Q87** (8)	**Q88** (4)
Q89 (17)	**Q90** (7)		